

Topic 1: What is Whole Exome Sequencing?

The human genome is roughly 3 billion base pairs long, but only about 1–2% of it encodes proteins. This protein-coding portion is called the **exome**, and it is estimated to harbour ~85% of disease-causing mutations. WES selectively captures and sequences only this portion, making it far more cost-effective than whole-genome sequencing when the goal is to find coding variants.

1. DNA is fragmented into small pieces (~150–300 bp).
2. Biotinylated RNA "baits" hybridize to exonic regions.
3. Streptavidin-coated beads pull down the captured fragments.
4. The enriched library is sequenced on an Illumina platform (typically paired-end 150 bp).
- 5.

Metric	What it means	Target
On-target rate	% of reads mapping to exome baits	>70%
Mean target coverage	Average depth across exonic bases	≥50× (germline), ≥100× (somatic)
% bases at ≥20×	Fraction of exome with adequate depth	>90%
Uniformity	Even coverage across targets	Coefficient of variation <0.5

Topic 2: Workflow Overview

```
FASTQ (raw reads)
  ↓ fastp (QC + trimming)
Trimmed FASTQ
  ↓ BWA-MEM (alignment)
SAM file
  ↓ samtools view/sort (convert + sort)
Sorted BAM
  ↓ Picard MarkDuplicates
Duplicate-marked BAM
  ↓ GATK BaseRecalibrator + ApplyBQSR
Recalibrated BAM
  ↓ GATK HaplotypeCaller
GVCF → VCF
  ↓ VQSR / hard filters
Filtered VCF
  ↓ VEP / ANNOVAR
Annotated VCF / MAF
```

Topic 3: Quality Assessment Metrics

Before any analysis, you must verify that the raw sequencing data is trustworthy. Sequencing machines produce FASTQ files where each base call is accompanied by a **Phred quality score (Q score)** -a log-scaled measure of the probability that the base call is wrong.

Q20 = 1% error probability Q30 = 0.1% error probability

Most downstream tools require reads to be predominantly Q30.

Tool: FastQC / fastp

Metric	Good sign	Warning sign
Per-base quality score	Q30 across the full read length	Drop below Q20 at read ends
Per-sequence quality score	Peak at Q35–40	Broad low-quality peak
GC content	Smooth bell curve around expected GC%	Bimodal or shifted curve
Sequence duplication	<20% for WES	>50% suggests library complexity issues
Adapter content	Near zero after trimming	Untrimmed adapters contaminate alignment
N content	<1%	High N content = sequencer flow cell problems

What to do about bad quality

- Trim low-quality bases from read ends (fastp --cut_tail, --qualified_quality_phred 20)
- Remove reads shorter than 50 bp after trimming
- Flag runs with >30% duplication for potential re-sequencing

Topic 4: Reference Genomes

A reference genome is a consensus representation of the human genome assembled from multiple individuals. The two primary builds used in human genomics are **GRCh37 (hg19)** and **GRCh38 (hg38)**. For new projects, hg38 is strongly preferred -it has better representation of centromeres, telomeres, and alternative loci, and all major databases (gnomAD, ClinVar, dbSNP) have hg38-native builds.

Critical points:

- Chromosome naming differs between sources: UCSC uses chr1, chr2 etc.; Ensembl/NCBI uses 1, 2 etc. This causes silent failures in pipelines if not matched consistently.
- The reference must be indexed before use (.fai for samtools, .dict for GATK, BWA index files .bwt, .sa, .ann, .amb, .pac).
- Always include **decoy sequences** (hs38d1) and **ALT contigs** in your reference. Reads that would otherwise mismatch to the primary assembly get correctly captured by decoys, improving alignment specificity.

Indexing commands

```
# BWA index
bwa index -a bwtsw GRCh38_full_analysis_set_plus_decoy_hla.fa

# samtools fai
samtools faidx GRCh38_full_analysis_set_plus_decoy_hla.fa

# GATK sequence dictionary
gatk CreateSequenceDictionary -R GRCh38_full_analysis_set_plus_decoy_hla.fa
```

Topic 5: BWA-MEM Algorithm

BWA-MEM (Burrows-Wheeler Aligner -Maximal Exact Matches) is the standard short-read aligner for Illumina WES/WGS data. It uses the Burrows-Wheeler Transform (BWT) to index the reference into a compressed data structure, enabling extremely fast lookup.

The algorithm works in three stages:

1. **Seeding** -finds maximal exact matches (MEMs) between the read and the reference index. These are short anchor sequences that match perfectly.
2. **Chaining** -groups nearby seeds that are consistent with a linear alignment.
3. **Extension** -uses Smith-Waterman dynamic programming to fill in gaps between seeds and extend to the full alignment, allowing mismatches and indels.

Standard command

```
bwa mem -t 16 -R "@RG\tID:sample1\tSM:sample1\tPL:ILLUMINA\tLB:lib1" \
reference.fa \
sample_R1.fastq.gz sample_R2.fastq.gz \
| samtools sort -@ 8 -o sample.sorted.bam
```

Key parameter: Read Groups (-R)

The @RG tag is mandatory for GATK. It embeds sample identity, platform, and library information into the BAM. Without it, GATK HaplotypeCaller will refuse to run. Every BAM must have **SM**: (sample name), **PL**: (platform), **LB**: (library), and **ID**: (read group ID) populated correctly.

Key metrics after alignment (samtools flagstat)

Metric	Expected
% mapped reads	>95% for WES from human samples
% properly paired	>90%
% singletons	<5%

Topic 6: SAM/BAM Formats

SAM (Sequence Alignment/Map) is a plain-text tab-delimited format storing each read's alignment information. **BAM** is the binary compressed equivalent -same data, roughly 4–5× smaller, and queryable with random access via an index (.bai).

A SAM file has two sections:

Header (@ lines): reference sequences (@SQ), read groups (@RG), program records (@PG).

Alignment records: 11 mandatory fields per read.

The 11 mandatory SAM fields:

Field	Name	Example	Meaning
1	QNAME	read001	Read name
2	FLAG	99	Bitwise alignment flags
3	RNAME	chr7	Reference chromosome
4	POS	117548628	1-based leftmost mapping position
5	MAPQ	60	Mapping quality
6	CIGAR	101M	Alignment description
7	RNEXT	=	Mate's chromosome (= if same)
8	PNEXT	117548750	Mate's position
9	TLEN	223	Template/insert length
10	SEQ	ACGT...	Read sequence
11	QUAL	IIII...	Base quality (Phred+33 ASCII)

The FLAG field

FLAG is a bitwise integer. Each bit represents a yes/no property. Common values:

FLAG	Meaning
0x1 (1)	Read is paired
0x2 (2)	Read mapped in proper pair
0x4 (4)	Read is unmapped
0x8 (8)	Mate is unmapped
0x10 (16)	Read is on reverse strand
0x100 (256)	Secondary alignment
0x400 (1024)	PCR or optical duplicate
0x800 (2048)	Supplementary alignment

A FLAG of 99 = 1+2+32+64 = paired, proper pair, mate on reverse strand, read 1.

Use `samtools flags 99` to decode any FLAG value.

The CIGAR string

CIGAR describes how the read aligns:

Op	Meaning
M	Match or mismatch
I	Insertion relative to reference
D	Deletion relative to reference
S	Soft clip (bases present but not aligned)
H	Hard clip (bases not present in SEQ)

Op	Meaning
N	Skipped region (intron in RNA-seq)

Example: 10S91M = first 10 bases soft-clipped (adapter), then 91 bases aligned.

Conversion and sorting commands

```
# SAM to BAM
samtools view -bS -@ 8 sample.sam -o sample.bam

# Sort by coordinate (required for GATK)
samtools sort -@ 8 sample.bam -o sample.sorted.bam

# Index
samtools index sample.sorted.bam

# Quick stats
samtools flagstat sample.sorted.bam
samtools idxstats sample.sorted.bam
```

Topic 7: Mapping Quality (MAPQ)

MAPQ (Mapping Quality, field 5 in SAM) is a Phred-scaled probability that the read is mapped to the **wrong** location. It is calculated by the aligner based on how uniquely and well the read maps.

MAPQ 60 = probability of wrong mapping is 10^{-6} (best possible in BWA-MEM) MAPQ 0 = read maps equally well to multiple locations (multimapper) MAPQ 1–10 = low confidence mapping

- GATK HaplotypeCaller **ignores reads with MAPQ < 20** by default
- Low MAPQ reads in variant-called regions are a warning flag
- Regions with consistently low MAPQ (e.g., segmental duplications, pseudogenes) produce false positive variants

Check MAPQ distribution

```
samtools view -q 20 sample.sorted.bam | wc -l # reads with MAPQ>=20
samtools view sample.sorted.bam | awk '{print $5}' | sort -n | uniq -c # MAPQ histogram
```

Topic 8: Duplicate Reads

During library preparation, DNA fragments are amplified by PCR. If the same original DNA molecule is amplified many times and sequenced multiple times, those copies are **PCR duplicates**. They carry identical sequence and start/end positions, making them identifiable.

Optical duplicates are a related artifact -they arise from the sequencer misidentifying one cluster as two. Both types must be flagged and excluded from variant calling, because counting duplicates inflates apparent read depth and makes variant allele frequencies (VAFs) look higher or lower than they are.

Tool: Picard MarkDuplicates / GATK MarkDuplicatesSpark

The tool identifies read pairs with the same 5' start position on both strands and flags them with FLAG 1024 (0x400). It marks -not removes -duplicates by default. Downstream tools then ignore flagged reads.

```
gatk MarkDuplicatesSpark \  
-I sample.sorted.bam \  
-O sample.markdup.bam \  
-M sample.markdup.metrics.txt \  
--tmp-dir /tmp
```

Metric	Meaning	Concern threshold
PERCENT_DUPLICATION	% of reads marked as duplicates	>30% for WES
ESTIMATED_LIBRARY_SIZE	Estimated unique molecules in library	Low = poor library complexity
READ_PAIR_DUPLICATES	PCR duplicate pairs	Main duplication source
READ_PAIR_OPTICAL_DUPLICATES	Sequencer-derived duplicates	High if optical distance too large

Topic 9: GATK Best Practices

The GATK Best Practices pipeline (developed by the Broad Institute) is the community standard for germline short variant discovery from Illumina data. It prescribes a specific sequence of pre-processing steps before variant calling to minimise systematic errors introduced by sequencing chemistry and library preparation.

The two pre-processing steps specific to GATK are:

1. Base Quality Score Recalibration (BQSR)

The quality scores assigned by the sequencer are empirically imprecise -they are derived from a signal model, not from actual error rates. BQSR builds a statistical model of actual error rates by comparing base calls at known variant positions (from dbSNP) and known invariant sites. It then adjusts all quality scores in the BAM to match empirical reality.

Two-step process:

- **BaseRecalibrator** → builds recalibration table
- **ApplyBQSR** → writes a new BAM with adjusted Q scores

```
gatk BaseRecalibrator \  
-I sample.markdup.bam \  
-R reference.fa \  
--known-sites dbSNP_146.hg38.vcf.gz \  
--known-sites Mills_and_1000G_gold_standard.indels.hg38.vcf.gz \  
-O recal.table
```

```
gatk ApplyBQSR \  
-I sample.markdup.bam \  
-R reference.fa \  
--bqsr-recal-file recal.table \  
-O sample.recal.bam
```

2. HaplotypeCaller -local realignment

HaplotypeCaller performs **local de novo assembly** of reads in active regions. When it detects evidence of variation, it assembles a De Bruijn graph of all reads in that region, constructs candidate haplotypes, and then re-aligns reads to those haplotypes using pair-HMM before scoring genotypes.

Topic 10: Variant Discovery

Variant discovery is the process of identifying positions where sequenced reads differ from the reference genome. These differences can be:

- **SNV** (single nucleotide variant): one base differs from reference
- **Indel**: insertion or deletion of one or more bases
- **MNV** (multi-nucleotide variant): multiple adjacent substitutions

Variants are distinguished from sequencing errors by statistical modelling of base quality scores, mapping quality, read strand bias, and allele frequency.

For example, if you have 100 typed copies of a manuscript that was copied from an original. If 95 copies say "cat" and 5 say "car", you'd suspect those 5 are typing errors. But if 50 say "cat" and 50 say "car", you'd suspect the original had two different versions. Variant calling is exactly this - computing whether a deviation from the reference is statistically likely to be a true biological difference or just noise.

Genotype likelihood: The probability of observing the data given a particular genotype (e.g., 0/0, 0/1, 1/1). HaplotypeCaller computes likelihoods for all possible genotypes at every site and picks the most probable.

Allele Frequency (AF): The fraction of reads supporting the alternate allele. Heterozygous germline variants cluster around 0.5; homozygous alts cluster near 1.0.

Genotype Quality (GQ): Phred-scaled confidence in the genotype call. $GQ \geq 20$ is usually required for reliable calls.

Topic 11: GVCF Workflow

The **GVCF** (genomic VCF) workflow is GATK's approach to joint genotyping across multiple samples. Rather than calling variants per sample independently, the GVCF workflow involves:

1. **Per-sample HaplotypeCaller** → produces a `.g.vcf` file containing a record for **every** position in the genome (not just variant sites), with genotype likelihoods stored even at non-variant sites.
2. **GenomicsDBImport** → imports all per-sample GVCFs into a database.
3. **GenotypeGVCFs** → performs joint genotyping across all samples simultaneously.

Why GVCF over direct multi-sample calling?

- You can add new samples without reprocessing old ones
- Joint genotyping dramatically improves variant detection at low-frequency sites
- Reference confidence model handles missing data correctly

```
# Step 1: per-sample GVCF
gatk HaplotypeCaller \
  -R reference.fa \
  -I sample.recal.bam \
  -O sample.g.vcf.gz \
  -ERC GVCF \
  --dbsnp dbSNP_146.hg38.vcf.gz

# Step 2: combine GVCFs
gatk GenomicsDBImport \
  -V sample1.g.vcf.gz \
  -V sample2.g.vcf.gz \
  --genomicsdb-workspace-path cohort_db \
  -L intervals.bed

# Step 3: joint genotyping
gatk GenotypeGVCFs \
  -R reference.fa \
  -V gendb://cohort_db \
  -O cohort.vcf.gz
```

Topic 12: Variant Filtration

Raw VCF output from HaplotypeCaller contains both true variants and false positives driven by systematic errors. Filtration removes or flags unreliable calls. There are two strategies:

VQSR (Variant Quality Score Recalibration) -preferred for large cohorts (>30 WES samples)

Builds a Gaussian mixture model using known variant truth sets (HapMap, 1000G, dbSNP, Mills indels) to score each variant's likelihood of being real. Variants are filtered by a VQSLOD (log-odds) threshold.

```
# SNPs
gatk VariantRecalibrator \
  -V cohort.vcf.gz \
  --resource:hapmap,known=false,training=true,truth=true,prior=15
hapmap.vcf.gz \
  --resource:dbsnp,known=true,training=false,truth=false,prior=2
dbsnp.vcf.gz \
  -an QD -an MQ -an MQRankSum -an ReadPosRankSum -an FS -an SOR \
  -mode SNP \
  -O snp.recal \
  --tranches-file snp.tranches

gatk ApplyVQSR \
  -V cohort.vcf.gz \
  --recal-file snp.recal \
  --tranches-file snp.tranches \
  --truth-sensitivity-filter-level 99.5 \
  -mode SNP \
  -O filtered.vcf.gz
```


Hard filtration -for small cohorts or single samples

Apply fixed thresholds on annotation fields. GATK recommended hard filters for SNPs:

Filter	Expression	Rationale
QD < 2.0	QualByDepth	Low quality relative to depth
FS > 60	FisherStrand	Strand bias
MQ < 40	MappingQuality	Poor average mapping quality
MQRankSum < -12.5	MappingQualityRankSum	Alt reads have worse MAPQ
ReadPosRankSum < -8.0	ReadPositionRankSum	Alt allele at read ends
SOR > 3.0	StrandOddsRatio	Strand imbalance

For indels: QD < 2.0, FS > 200, ReadPosRankSum < -20.0.

Topic 13: VCF Format

The **VCF** (Variant Call Format) is the standard file format for storing variants. It is a tab-delimited plain-text file with a header section and one line per variant record.

VCF header lines start with ## and define the format version, reference, contigs, INFO fields, FORMAT fields, and FILTER definitions. The last header line starts with #CHROM and lists the column names.

The 8 fixed columns + FORMAT + sample columns

Column	Name	Example	Meaning
1	CHROM	chr7	Chromosome
2	POS	117548628	1-based position
3	ID	rs121913527	dbSNP ID or .
4	REF	A	Reference allele
5	ALT	T	Alternate allele(s)
6	QUAL	3842.77	Phred-scaled variant quality
7	FILTER	PASS	Filter status
8	INFO	DP=187;AF=0.52	Annotation key-value pairs
9	FORMAT	GT:AD:DP:GQ:PL	Format of sample columns
10+	SAMPLE	0/1:89,98:187:99:3871,0,3421	Per-sample genotype data

FORMAT: GT:AD:DP:GQ:PL Sample: 0/1:89,98:187:99:3871,0,3421

- GT = 0/1 → heterozygous (ref/alt)
- AD = 89,98 → 89 ref reads, 98 alt reads
- DP = 187 → total depth
- GQ = 99 → genotype quality (capped at 99)
- PL = 3871,0,3421 → phred-scaled likelihoods for 0/0, 0/1, 1/1 (0 = most likely)

Key INFO fields

Field	Meaning	Good value
DP	Total read depth at site	≥20 for WES
AF	Allele frequency	~0.5 (het), ~1.0 (hom)
QD	Quality by depth	>2.0
FS	Fisher strand bias	<60 for SNPs
MQ	Mean mapping quality	>40
VQSLOD	VQSR log-odds score	>0 (positive = passes)

Topic 14: Variant Annotation

A raw VCF tells you *where* a variant is and *what* the change is and annotation adds biological and clinical context.

- **Gene/transcript context:** which gene, which exon, what codon, what protein change
- **Consequence:** synonymous, missense, stop-gain, frameshift, splice-site, etc.
- **Population frequency:** is this variant common (likely benign) or rare (possibly pathogenic)?
- **Functional prediction:** SIFT, PolyPhen-2, CADD score -is the amino acid change predicted to damage protein function?
- **Clinical databases:** ClinVar pathogenicity assertion, COSMIC somatic mutation records
- **Conservation scores:** GERP, PhyloP -is this position conserved across species?

Standard tools

- **VEP** (Variant Effect Predictor, Ensembl): comprehensive, plugin-extensible, outputs per-transcript consequences
- **ANNOVAR:** tabular, fast, good for population databases
- **SnpEff:** fast annotation with own database build

Key annotation fields

Annotation	Source	Why it matters
Consequence	VEP	Primary functional classification
SYMBOL / Gene	VEP	Gene name
HGVSc / HGVSnp	VEP	Standard variant nomenclature
gnomAD_AF	gnomAD	Population frequency (filter if >1%)
CADD_PHRED	CADD	Combined deleteriousness score (>20 = top 1% most deleterious)
ClinVar_CLNSIG	ClinVar	Clinical significance
PolyPhen / SIFT	VEP	Protein function prediction
IMPACT	VEP	HIGH/MODERATE/LOW/MODIFIER

IGV Visualisation

Load in this order in IGV:

1. File → Load Genome → GRCh38 (or load local .fa + .fai)
2. File → Load from File → patient01.markdup.bam
3. Navigate to a known variant region (e.g., BRCA1: chr17:43,044,295)

Feature	Good	Concern
Read coverage	Consistent depth across exon	Sudden drops or gaps
Read colour (strand)	Mix of red/blue reads	All reads one colour (strand bias)
Base mismatches	Rare, scattered	Clustered at read ends
Paired-end orientation	Reads pointing inward	Reads pointing same direction
Soft clips	Minimal	Extensive clipping = alignment artifact

Quick VCF inspection:

```
# Count total variants
bcftools stats patient01_filtered.vcf.gz | grep "number of SNPs"

# Look at PASS variants only
bcftools view -f PASS patient01_filtered.vcf.gz | bcftools stats | grep "^SN"

# Ti/Tv ratio -critical QC metric
bcftools stats patient01_filtered.vcf.gz | grep "Ts/Tv"

# View specific variant in text
bcftools view patient01_filtered.vcf.gz chr17:43044295-43044295
```

Ti/Tv ratio -the most important post-calling QC metric

Transitions (Ti): purine ↔ purine (A ↔ G) or pyrimidine ↔ pyrimidine (C ↔ T) -chemically more likely
Transversions (Tv): purine ↔ pyrimidine -less likely

Expected Ti/Tv ratios:

- Whole genome: ~2.0–2.1
- Whole exome: ~2.8–3.0 (exons are under stronger selection pressure, enriching Ti)
- Below 2.5 for WES = excess false positives (artifactual transversions)
- Above 3.5 for WES = possible reference bias or filtering over-restriction

Key Metrics

Step	Tool	Key Metric	Target
Raw QC	FastQC/fastp	% Q30 bases	>80%
Alignment	BWA-MEM + samtools	% mapped reads	>95%
Alignment	samtools flagstat	% properly paired	>90%
Coverage	Picard/GATK	Mean target depth	≥50×
Coverage	Picard/GATK	% bases ≥20×	>90%
Duplication	MarkDuplicates	% duplicates	<20%
MAPQ	samtools	% reads MAPQ≥20	>90%
Variant QC	bcftools stats	Ti/Tv ratio	2.8–3.0 (WES)
Genotype	VCF	GQ per call	≥20
Filtration	GATK	VQSLOD score	>0 (PASS)